

## EDUCATION | Toronto Metropolitan University

- Bachelor of Engineering, Mechatronics Engineering
  - Expected Graduation in 2029
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## EXPERIENCE

### Undergraduate Research Assistant | Toronto Metropolitan University

- Contributed hardware development at the ELIXIR Lab under Dr. Robin Chhabra, a research group advancing embodied AI and soft robotics for real-world autonomous systems including assistive, aerial, and space-capable robots.
- Conducted research under Dr. Chhabra, applying iterative **PCB** design reviews, signal integrity considerations, and hardware debugging methodologies to meet experimental research requirements
- Developed custom PCB designs for soft robotics research platforms with integrated power management, sensor interface circuits, and actuation control systems for next-generation flexible robotic applications

### Electrical Engineering Lead — Engineers for a Sustainable World | Toronto Metropolitan University

- Led embedded and electrical systems design for a funded algae photobioreactor carbon capture system, using **ESP32/Raspberry Pi** with real-time sensor monitoring, automated control algorithms, and integrated data acquisition architecture
  - Built and deployed a multi-sensor data acquisition and visualization system using multiple **PCBs** to monitor algae growth rates, CO<sub>2</sub> uptake efficiency, temperature, pH, and humidity, integrating real-time data logging, signal conditioning, and dashboard-based performance analytics for continuous system optimization
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## PROJECTS

### Soft Robot Controller | PCB Design, Circuit Design, KiCAD, ESP32, Arduino, Python, C++, Control Systems

- Engineered a **4-layer** custom control PCB on KiCAD for the Elixir Labs soft robotics research platform at TMU, consolidating power regulation, sensing, and actuation across **9** components onto a single board.
- Integrated DRV8874 motor drivers with an ESP32-S3, balancing noise isolation and current capacity across motor drive and sensor signal paths to support multi-channel actuation control for soft robotic end-effectors.
- Implemented a multi-rail power architecture spanning **4** voltage levels (7.6V, 3.3V, 2.8V, 1.2V) using a SYB113A0C buck converter and dual LDO regulators, supplying 2 DRV8874 motor controllers with **3A** lockout protection alongside an ESP32-S3, LSM6DSOXTR IMU, and camera interface.
- Implemented iterative firmware testing and CI-style debugging, reducing board rework cycles and improving prototype stability.

### Haptic Feedback Watch | PCB Design, Circuit Design, KiCAD, ESP32, Arduino, Python, C++

- Designed a compact **2-layer** wrist-worn PCB on KiCAD centered on an ESP32-C3, integrating haptic feedback actuation circuitry to deliver real-time fitness alerts within the tight form factor constraints of a wearable device.
- Produced fabrication-ready Gerber files and a complete BOM across **5** components, iterating on placement and trace routing to meet wearable size requirements while maintaining full electrical functionality.
- Designed a complete power subsystem incorporating a USB-C input, MCP73831 LiPo charge controller, and AMS1117 LDO regulator to deliver a stable **3.3V** rail across the ESP32-C3, LIS3DHTR accelerometer, and DRV8837 haptic driver.

### Harmoniq - Modular Music Generation | ESP32, ElevenLabs API, Arduino IDE, C++, Soldering, Circuit Design

- Built an **ESP32**-based modular music system detecting **3** physical magnetic instrument blocks (piano, guitar, drums) that dynamically trigger ElevenLabs API calls to regenerate backtracks in real time.
  - Soldered and assembled the full hardware prototype in under **24** hours at MakeMIT x Harvard Hackathon, contributing to block detection circuitry and ESP32 firmware alongside a team of **4** engineers.
  - Optimized magnet alignment configurations to ensure consistent block detection for users aged 4–7.
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## TECHNICAL SKILLS

- **PCB & Electrical Design:** PCB Design, Schematic Capture, Layout Optimization, Power Plane Design, KiCAD, Altium
- **Power & Hardware:** Buck Converters, LDO Regulators, LiPo Charging, USB-C Power, Motor Drivers, Power Management
- **Embedded Systems:** ESP32, STM32, Arduino, UART, SPI, I2C, Timers, RTC, Sensor Interfacing, Actuator Control, C, C++
- **Hardware Debugging & Validation:** Oscilloscope, Multimeter, Signal Probing, Power Rail Verification, Prototyping, Soldering
- **Developmental Tools:** Git, GitHub, Arduino IDE, Serial Debugging, Firmware Integration